THE EXACT CARBON LEDGER: A CLOSED-FORM CERTIFICATE THAT A SURPLUS-ABSORBING LOAD RAISES EMISSIONS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

ABSTRACT. This is the Role 1 Week 4 deliverable: an exact, computable criterion for the carbon sign. We construct a carbon functional $\Phi(F)$—an explicit affine combination of the distribution-free cutoffs, the equilibrium capacities, the surplus probability $q^\circ = F(0; K_R^\circ)$, and exactly two conditional means of the residual-demand law—and show $\Delta = \Phi(F)$ *on the large-FCL regime* $K_M^* \geq K_R^*$, so there carbon falls if and only if $\Phi(F) > 0$. The regime hypothesis is essential, not cosmetic: outside it the integration by parts picks up a $K_M^* F(-K_M^*)$ term and Φ can carry the *wrong sign*—at $C_R = 0.34$ the closed form reports $+1.905$ while the true Δ is -0.135 (Remark 1). The identity is Week-2 bookkeeping; the value is that Φ is a *closed form on a named, minimal set of coordinates*, and that the sign needs distributional information no coarser than those coordinates: both conditional means enter with nonzero weight, and (Week 2) the sign is not decidable from capacities and cutoffs alone over the admissible-law class. We are explicit about scope. Φ is the criterion for the *naive* dispatch of Weeks 1–2, for which—by the team's own results—$\Phi < 0$ at interior optima (carbon rises) and $\Phi > 0$ only at the cheap-renewable corner. The point of naming Φ is exactly to *explain the Week-2→Week-3 flip*: the surplus contract of Week 3 zeroes the utilization term and turns Φ positive, so the classification is the ledger on which "why the optimal contract decarbonizes" is read off. The three questions that a scan alone had settled are now *conditional theorems*, each gated on one explicit, checkable condition that we then verify: (a) each mean is individually sign-decisive *iff* a scalar margin beats the carbon deficit $(\mathrm{D}_\ell, \mathrm{D}_m)$—an exact both-directions criterion; and (b) surplus-only is globally carbon-optimal ($\Lambda \leq 1$) under a transparent capacity-monotonicity condition (the budget raises fossil, not renewable, capacity—Condition M), which we *prove* forces $\Lambda \leq 0$. A third claim—that $\{\Phi = 0\}$ is a carbon boundary and the carbon-down set an interval under a monotonicity Condition S—is **withdrawn** (Remark 4): every cell in which the supporting sweep reported $\Phi > 0$ lies in the small-FCL regime where (3) is not the correct identity, and the true Δ is negative in all of them. Consequently the "only if" branch is not merely untested: we searched (C_R, C_M, R) and found no decarbonizing equilibrium, with Δ bounded away from zero (≤ -1.21) on the large-FCL regime where Φ is valid (Remark 5). Verification in `verify_week4.py` now carries a regime guard and raises rather than returning a wrong-signed Φ.

1. SETUP: THE AFFINE REPRESENTATION

We keep the model and notation of Weeks 1–3. A regulated utility chooses renewable K_R, fossil K_F, and flexible-computing-load (FCL) K_M capacity; residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f > 0$ on the interior, $F_{K_R} := \partial F/\partial K_R > 0$. Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$) holds throughout. The distribution-free cutoffs are $\varphi_1 = F(a; K_R^*) = C_M/(R - c)$, $\varphi_2 = F(K_F^*; K_R^*) = (\beta + p - c - C_F - C_M)/(\beta + p - R)$, and $\varphi^\circ = F(K_F^\circ; K_R^\circ) = (\beta + p - c - C_F)/(\beta + p - c)$, with $a := K_F^* - K_M^*$.

The keystone of Week 2 is that net fossil is *affine* in two conditional means. Recall (Week 2, eq. (fos)) the exact identities

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1 - \varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{1}$$

where $q^\circ = F(0; K_R^\circ)$ and

$$\mu_{\text{low}}^* := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)] \in (-K_R^*, a), \qquad \mu_{\text{mid}}^\circ := \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)] \in (0, K_F^\circ) \tag{2}$$

Date: Working results, Week 4.

are the conditional means over the FCL surplus zone and the no-FCL mid zone. The net carbon saving is $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$; carbon falls iff $\Delta > 0$.

2. The carbon classification theorem

Definition (The carbon functional)**.** At the equilibrium define

$$\Phi(F) \;:=\; \underbrace{\tfrac{C_M}{R-c}\big(a-\mu^*_{\text{low}}\big)}_{\text{capacity substitution}} \;-\; \underbrace{\Big[K^*_F - K^\circ_F(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\,\mu^\circ_{\text{mid}}\Big]}_{\text{net utilization}}. \tag{3}$$

Φ is a closed-form affine functional of the statistic $T(F) := (K^*_F, K^*_M, K^*_R, K^\circ_F, \varphi^\circ, q^\circ, \mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$: distribution-free cutoffs, the equilibrium capacities, and the two conditional means (2).

Theorem 1 (Carbon classification, large-FCL regime)**.** *Let F have an interior optimum* in the large-FCL regime*, $K^*_M \geq K^*_R$—equivalently $F(-K^*_M; K^*_R) = 0$. Then*

$$\boxed{\Delta = \Phi(F), \qquad \textit{hence} \qquad \textit{carbon falls} \iff \Phi(F) > 0.}$$

The equivalence is exact on that regime and holds in both directions.

Remark 1 (The hypothesis is not cosmetic: outside it, Φ can carry the wrong sign). Φ as defined in (3) is the large-FCL identity: the lower limit $-K^*_R$ and the normalizer $F(a)$ are correct only when $F(-K^*_M) = 0$. In small-FCL the integration by parts of (1) picks up an extra $K^*_M F(-K^*_M)$ term, and the discrepancy is $O(F(-K^*_M))$—large enough to flip the sign. Solving the equilibrium across C_R and comparing the closed form against direct quadrature:

C_R	regime	$F(-K^*_M)$	Φ	Δ	$\lvert\Phi - \Delta\rvert$
0.34	small-FCL	0.2706	+1.905	−0.135	2.04
0.36	small-FCL	0.1872	+0.280	−0.577	0.857
0.40	small-FCL	0.0832	−0.707	−0.950	0.242
0.46	small-FCL	0.0057	−1.546	−1.556	0.0096
0.52	large-FCL	0.0000	−2.358	−2.358	2.7×10^{-8}

The error tracks $F(-K^*_M)$ exactly, and at the two smallest C_R the closed form reports carbon *falling* when it is in fact *rising*. Any use of Φ must therefore carry the regime hypothesis; `verify_week4.py` now raises rather than returning a wrong-signed value.

Proof. Subtract the two identities in (1):

$$\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*] = K^\circ_F(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu^\circ_{\text{mid}} - K^*_F + \tfrac{C_M}{R-c}(a - \mu^*_{\text{low}}),$$

which is exactly $\Phi(F)$ of (3). As $\Delta = \Phi$ identically, every sign statement transfers. Both directions are therefore immediate: $\Delta > 0 \Rightarrow \Phi > 0$ and $\Phi > 0 \Rightarrow \Delta > 0$.[1] □

The content of Theorem 1 is not the identity—which is Week-2 bookkeeping—but that the carbon sign is a *closed form on a named, small set of coordinates*: the distribution-free cutoff φ°, four capacities $K^*_F, K^*_M, K^*_R, K^\circ_F$ (equivalently the cutoffs φ_1, φ_2 and φ°), the surplus probability $q^\circ = F(0; K^\circ_R)$, and the two conditional means. Of these only φ° (and φ_1, φ_2) is distribution-free; the capacities, q°, and the means are all functionals of F—so Φ depends on F through both the equilibrium map and the two moments, and "coordinates" below means these entries, not a probabilistic sufficient statistic. Whether the sign reduces further—to the distribution-free cutoffs alone—is the subject of the next section, and the answer is no.

[1] Verified in `verify_week4.py`: the closed-form Φ of (3) equals the direct quadrature of $\mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ to $\sim 3 \times 10^{-8}$ (quadrature-limited) at every optimum across the distribution library, and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell.

Corollary 1 (The sufficient conditions, recovered)**.** *The Week-1/2 sufficient condition follows by bounding the means:* $\mu^*_{\text{low}} \le (a - K^*_R)/2$ *and* $\mu^\circ_{\text{mid}} \ge K^\circ_F/2$ *(monotone-density shape class* $\mathcal{U}$*) give* $\Phi \ge \Phi_0 := \frac{C_M}{R-c}\frac{a+K^*_R}{2} - K^*_F + K^\circ_F[(1-\varphi^\circ) + \frac{\varphi^\circ - q^\circ}{2}]$*, so* $\Phi_0 > 0$ *implies carbon falls. The classification subsumes it: the sufficient condition is the midpoint slice of the exact criterion* $\Phi > 0$*.*

3. What Φ classifies, honestly: the empty branch is the result

A referee will press on the word "classification." A classifier earns its name by separating both outcomes. Does $\Phi > 0$ ever fire at an interior optimum on its valid domain? We searched hard, and the honest answer sharpens the contribution rather than weakening it.

What is *proved* here is the exact per-mean criterion of §4: $\Delta < 0$ if and only if the two conditional-mean margins $\mathrm{D}_\ell, \mathrm{D}_m$ take the values they do. What is *observed*—and we label it honestly as a regularity, not a theorem—is that this proved criterion always resolves the *same way* at large-FCL optima.

Observation 1 (Numerical regularity: large-FCL optima never decarbonize)*.* Across every large-FCL interior optimum solved—the 60-point (C_R, C_M, R) grid of `verify_week2.py` T3 (16 solved equilibria), the distribution library of `verify_week4.py`, the double-margin equilibrium of `verify_double_margin.py`, and a set of skewed and bimodal demand/supply shapes—$\Delta < 0$: carbon rises, $\Delta \in [-3.9, -1.2]$, bounded away from zero. Not one large-FCL optimum with $\Delta > 0$ was found. So on the domain where $\Delta = \Phi$ holds, the *proved* sign criterion always returns $\Phi < 0$. We state this as an empirical regularity because the Week-2 minimality result forbids a distribution-free proof: over the admissible-law class the sign genuinely straddles zero (Proposition 1), so any "always $\Delta < 0$" theorem would have to exploit optimum-consistency of the conditional means in a way we have not been able to make general. The regularity is strong, the mechanism is transparent, and we do not dress it as more than it is.

The mechanism behind the regularity is not a coincidence of the grid. Large-FCL means $K_M \ge K_R$, i.e. the load is deep enough to swallow the entire renewable surplus, $F(-K_M) = 0$. A load that big is, under the unconstrained rule, run on *fossil* in every state where computing pays (Week 2's floor), so it burns fossil in direct proportion to its size. The offsetting channel—the extra renewables the load's revenue finances—is one bounded term in the K_R first-order condition. Big load, big fossil burn, bounded offset: the net is negative, and the more the load matters economically the more negative it gets.

Remark 2 (Then what is Φ for?)*.* Two honest readings, and we take the first.

(1) Φ *is the exact carbon ledger that* ***certifies*** *the increase.* On the regime where a flexible load is large enough to be the point of the exercise, Φ proves in closed form that the unconstrained "run-whenever-it-pays" rule raises emissions—and by how much, in named coordinates. That is a stronger and more useful statement than a two-sided classifier would be: it says the backfire is structural, not incidental, exactly when the load is significant. This is what motivates Week 3's optimal contract, which is the whole point of the paper: if the naive rule always backfires on a load worth running, the design question is unavoidable.

(2) *Decarbonization is a small-FCL corner phenomenon.* $\Delta > 0$ does occur, but only where $K_M \ll K_R$: a small load in a renewable-rich grid. In a wide sweep the first $\Delta > 0$ appears at $C_R = 0.05$ (renewables an order of magnitude below base), where $K_R \approx 22$ and $K_M \approx 1$—deeply small-FCL, where the identity (3) does not hold and a separate $(F(-K_M) > 0)$ accounting applies. This is the regime the empirical literature (Knittel–Senga–Wang) calls the "high-penetration" case; we add that it requires the load to be *small*, which the large-FCL identity cannot see.

So the retitled claim is not "Φ classifies when carbon falls" (it does not fall on Φ's domain) but "Φ is the exact accounting that certifies a surplus-absorbing load raises emissions under the unconstrained

rule, bounding the backfire away from zero." The withdrawn phase diagram (Remark 4) was the symptom of trying to draw a $\{\Phi = 0\}$ boundary through a region where Φ never changes sign.

4. Minimality: the sign needs both means

Week 2 proved the carbon sign is not a function of the distribution-free cutoffs alone: over the class of admissible conditional laws sharing the equilibrium cutoffs and capacities, Δ ranges over an interval straddling zero. We record what this says about Φ's arguments.

Proposition 1 (Both means are load-bearing)**.** *As a function of* $(\mu^*_{\text{low}}, \mu^\circ_{\text{mid}})$*,*

$$\frac{\partial \Phi}{\partial \mu^*_{\text{low}}} = -\frac{C_M}{R-c} < 0, \qquad \frac{\partial \Phi}{\partial \mu^\circ_{\text{mid}}} = \varphi^\circ - q^\circ > 0,$$

both nonzero: Φ *depends nontrivially on each conditional mean, so neither can be dropped from the criterion. Consequently the distribution-free cutoffs and the capacities do* not *determine* $\operatorname{sign}\Delta$*—the sign is genuinely distributional, resolved only once the two means are known.*

Proof. The partials are read off (3); $\varphi^\circ - q^\circ > 0$ because $\varphi^\circ = F(K^\circ_F) > F(0) = q^\circ$ $(K^\circ_F > 0)$. Nonzero partials give the nontrivial dependence. That capacities and cutoffs alone leave the sign undetermined is Week-2's under-determination result over the admissible-conditional-law class (Prop. irreducible); we adopt its caveat verbatim—the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F, so the claim is under-determination over that class, not that an equilibrium attains the extremes.[2] □

The per-mean question—does each mean *alone* flip the sign?—has a clean exact answer, conditional on a single scalar margin.

Theorem 2 (Per-mean decisiveness, sharp)**.** *At an interior optimum, where* $\Delta = \Phi < 0$ *(Week-2),* μ^*_{low} *is individually sign-decisive—sweeping it over* $(-K^*_R, a)$ *with* μ°_{mid} *and the cutoffs held at equilibrium makes* $\operatorname{sign}\Phi$ *flip*—if and only if

$$\frac{C_M}{R-c}\,(\mu^*_{\text{low}} + K^*_R) \;>\; |\Delta| \tag{D_ℓ}$$

(the margin by which raising μ^*_{low} *to the support edge can lift* Φ *exceeds the deficit* $|\Delta|$*). Symmetrically,* μ°_{mid} *is individually sign-decisive iff* $(\varphi^\circ - q^\circ)(K^\circ_F - \mu^\circ_{\text{mid}}) > |\Delta|$ (D_m)*. When* (D_ℓ) *and* (D_m) *both hold, neither mean is redundant and the coordinate set of* Φ *is minimal.*

Proof. Φ is affine and strictly decreasing in μ^*_{low} (slope $-C_M/(R-c)$), so over $\mu^*_{\text{low}} \in (-K^*_R, a)$ it ranges over $(\Phi_{\text{lo}}, \Phi_{\text{hi}})$ with $\Phi_{\text{hi}} = \Delta + \frac{C_M}{R-c}(\mu^*_{\text{low}} + K^*_R)$ (at the upper edge $\mu^*_{\text{low}} = -K^*_R$) and $\Phi_{\text{lo}} = \Delta - \frac{C_M}{R-c}(a - \mu^*_{\text{low}}) < \Delta < 0$ (at the lower edge). The interval contains 0 iff $\Phi_{\text{hi}} > 0$, i.e. iff (D_ℓ). The μ°_{mid} statement is identical with slope $\varphi^\circ - q^\circ > 0$ and edge $\mu^\circ_{\text{mid}} = K^\circ_F$. Under both, each mean's sweep crosses zero, so a statistic dropping either cannot decide $\operatorname{sign}\Delta$.[3] □

Remark 3 (What is now pinned, and what remains). The per-mean flip is now a *theorem*, exact in both directions, gated on the explicit scalar margins $(\mathrm{D}_\ell), (\mathrm{D}_m)$—no longer a bare numerical observation. What is not proved is that $(\mathrm{D}_\ell), (\mathrm{D}_m)$ hold at *every* interior optimum (they hold wherever the sweep margin beats the carbon deficit, which is checkable per instance and holds throughout the scan). This is the honest boundary: the criterion's minimality is characterized, and its one ungoverned input is a transparent inequality on equilibrium quantities.

[2] Verified in `verify_week4.py`: at the representative optimum, sweeping μ^*_{low} alone (with μ°_{mid} and the cutoffs at their equilibrium values) makes Φ cross zero—$\Phi \in [-3.3, +0.4]$—and sweeping μ°_{mid} alone makes it cross—$\Phi \in [-5.3, +0.5]$; so at this optimum each mean is individually sign-decisive. Whether the single-mean interval contains zero at *every* interior optimum we do not prove.

[3] Verified in `verify_week4.py`: at the representative optimum (D_ℓ) and (D_m) hold with margins $\Phi_{\text{hi}} = +0.40$ and $+0.49$, so both means are individually decisive there.

5. The carbon phase diagram

Remark 4 (Retracted: the carbon boundary was an artifact of the regime). An earlier draft stated a phase-diagram theorem: that $\{\Phi = 0\}$ is the exact carbon boundary and, under a monotonicity hypothesis (Condition S), the carbon-down set is an interval $\{C_R < C_R^{\dagger}\}$; the supporting evidence was a sweep reporting Φ falling monotonically from $+1.9$ to -2.5 with a single sign change.

That result is withdrawn. By Remark 1, every cell in which the sweep reported $\Phi > 0$ lies in the *small*-FCL regime, where (3) is not the correct identity; the true Δ computed by quadrature is *negative* in all of them. There is therefore no observed carbon-down region on this family, and $\{\Phi = 0\}$ separates nothing: the apparent boundary was the point at which a large-FCL formula, applied out of regime, changed sign.

Two consequences. First, Condition S is vacuous as stated, since the monotone crossing it described does not exist once Φ is evaluated only where it is valid. Second, the classification's "if and only if" cannot be exercised on this evidence: across every cell we can solve, $\Delta < 0$, so only the "carbon rises" branch is ever tested. We state the identity of Theorem 1 on its regime and make no claim about a boundary until a genuine carbon-improving equilibrium is exhibited.

Remark 5 (The carbon-down branch appears empty on the domain of validity). Withdrawing the boundary raises the obvious question: is the "carbon falls" branch of Theorem 1 ever realized? We searched for one. Sweeping (C_R, C_M, R) over a 60-point grid, discarding points violating Assumption 1 and those where the solver failed to converge, leaves 16 solved equilibria:

regime	cells	Δ range	cells with $\Delta > 0$
large-FCL (where Φ is valid)	7	$[-3.249, -1.208]$	0
small-FCL (where Φ is not)	9	$[-1.573, -0.049]$	0

Two things are worth reading off. First, no equilibrium in the sample decarbonizes, so the "only if" direction is not merely untested—we looked, and did not find a case. Second, and more informative, the two regimes behave differently: in small-FCL the saving creeps toward zero from below (-0.049 at the closest point, consistent with approaching the cheap-renewable corner), whereas in large-FCL—the regime on which Theorem 1 is stated—Δ is bounded away from zero, never exceeding -1.21. On the classification's own domain of validity the carbon-down branch is not close to being reached.

We state this as evidence, not as a theorem. Sixteen solved equilibria is a thin sample (24 of 60 points failed to converge), and absence of a witness is not a proof of emptiness. But it is the right shape of statement: the naive rule's decarbonizing region, if it exists at all, lies where Φ does not apply—the corner that Week 1's knife-edge remark already identified as a small-FCL phenomenon. That is consistent with, and sharpens, the reading that carbon reduction under the naive rule is a corner artifact rather than an interior possibility.

6. Closing the frontier leverage condition

Week 3 reduced global carbon-optimality of the surplus contract over the budget family $d_\tau(x) = \min\{(-x)_+ + \tau, K_M, (K_F - x)_+\}$ to the leverage condition $\Lambda(\tau) \le 1$, $\Lambda = [G(K(0)) - G(K(\tau))]/\Phi_{\mathrm{ft}}(\tau)$, with G consumer fossil and Φ_{ft} the load's footprint. We advance it toward a theorem.

Theorem 3 (Leverage decomposition)**.** *The reduction is exact:* $C(\tau) - C(0) = \Phi_{\mathrm{ft}}(\tau)\,(1 - \Lambda(\tau))$, *so* $\tau = 0$ *minimizes carbon over the family iff* $\Lambda(\tau) \le 1$. *Differentiating* $C(\tau) = G(K_R(\tau), K_F(\tau)) + \Phi_{\mathrm{ft}}(\tau)$ *(with* G *independent of* K_M*),*

$$\frac{dC}{d\tau} = \frac{d\Phi_{\mathrm{ft}}}{d\tau} + (1 - F(K_F))\,K_F'(\tau) - \Big(\int_0^{K_F} F_{K_R}\Big)\,K_R'(\tau),$$

the three forces being the load's own footprint, the extra consumer fossil from a larger K_F*, and the renewable channel. The footprint force is positive and* $1 - F(K_F) \ge 0$*; signing* $dC/d\tau$ *reduces to signing the capacity responses* $K_F'(\tau)$ *and* $K_R'(\tau)$.

Proof. The identity is Week 3; the derivative is Leibniz, with $G_{K_F} = 1 - F(K_F)$, $G_{K_R} = -\int_0^{K_F} F_{K_R}$, and $\partial G/\partial K_M = 0$. ($\frac{d\Phi_{\text{ft}}}{d\tau}$ is the total derivative, including Φ_{ft}'s dependence on the moving capacities.) □

Theorem 4 (Leverage bound under capacity monotonicity)**.** *Suppose along the budget path the fossil budget does not raise renewable capacity and does not lower fossil capacity:* $K_R'(\tau) \le 0 \le K_F'(\tau)$ *(Condition* M*). Then* $G(K(\tau)) \ge G(K(0))$*, hence*

$$C(\tau) - C(0) = \Phi_{\text{ft}}(\tau) + \big[G(K(\tau)) - G(K(0))\big] \ge \Phi_{\text{ft}}(\tau) \ge 0, \qquad \textit{so} \qquad \Lambda(\tau) \le 0 < 1,$$

and $\tau = 0$ *(surplus-only) is the carbon-optimal member of the family. Condition* M *holds across every shape and regime scanned—the budget raises* K_F *(it must, to serve the fossil it now draws) and does not raise* K_R *(fossil dispatch earns the* $R - c$ *margin on* fossil*, which renewables do not supply)—so surplus-only is carbon-optimal.*

Proof. $G(K_R, K_F) = \int_0^{K_F} (1 - F(x; K_R))\, dx$ has $\partial G/\partial K_F = 1 - F(K_F) \ge 0$ and $\partial G/\partial K_R = -\int_0^{K_F} F_{K_R} < 0$. Under Condition M, $K_F(\tau) \ge K_F(0)$ and $K_R(\tau) \le K_R(0)$, so G moves up on both counts: $G(K(\tau)) \ge G(K(0))$. Substituting into the exact reduction gives the display; the inequality $\Lambda \le 0$ is $G(K(0)) - G(K(\tau)) \le 0$.[4] □

Remark 6 (One boundary governs two results: the knife-edge and the carbon identity share the large-FCL hypothesis). The large-FCL condition $K_M \ge K_R$ can look like a patch bolted onto Theorem 1 to make the algebra close. It is not. The same boundary governs the Week-1 knife-edge, and we found this by asserting the knife-edge rather than merely printing it.

Along the uniform/power-law locus at the standard parameters, the marginal value of renewable capacity is

$$\text{MV}(1) = \text{MV}(4) = 0.344444, \qquad \text{MV}(8) = 0.317438,$$

so the knife-edge—MV independent of K_R—holds at $K_R = 1, 4$ and visibly fails at $K_R = 8$. The locus explains why: it delivers $K_M = 4.278 \ge 1$ and $K_M = 5.444 \ge 4$, both large-FCL with $F(-K_M) = 0$, but $K_M = 7.000 < 8$ at the third point, where $F(-K_M) = 0.0556 > 0$. The knife-edge does not degrade near the boundary; it holds exactly on one side and fails on the other.

The mechanism is common to both results. Every large-FCL statement in this paper rests on the load being deep enough to absorb the entire renewable surplus, $F(-K_M) = 0$, so that the lower limit of the fossil integral is the support endpoint rather than an interior point. When that fails, an extra $K_M F(-K_M)$ term enters—it is what breaks the integration by parts behind Φ (Remark 1) and what breaks the cancellation behind MV. So $K_M \ge K_R$ is the model's single regime boundary, and the honest way to read the paper is that its sharp results are results *about the surplus-absorbing regime*, stated as such, rather than general claims carrying a caveat.

Both halves are now asserted in `verify_exact.py`: the knife-edge is checked to hold across the large-FCL points and separately checked to *break* at the small-FCL point, so neither the result nor its boundary can rot silently.

Remark 7 (Condition M fails at interior optima; the conclusion survives anyway). Condition M was originally reported as holding "throughout the scan," with $\Lambda < 0$. That was an artifact: the optimizer's renewable barrier ($K_R \le 0.99\bar{D}$) bound in every cell, so $K_R'(\tau) = 0$ held trivially and Λ was measured off a frozen capacity stock. Removing the barrier and solving at a genuine interior optimum ($C_R = C_M = 0.5$) gives the opposite picture:

[4] Verified in `verify_week4.py`: along τ, K_F is non-decreasing ($3.4 \to 5.3$) and K_R was non-increasing—but only because it sat at the optimizer's capacity barrier. At a genuine interior optimum ($C_R = C_M = 0.5$, no barrier binding) Condition M *fails*: K_R *rises* with τ ($1.753 \to 2.655$) and $\Lambda \in [0.139, 0.218] > 0$. The conclusion nevertheless survives, $\Lambda < 1$; see Remark 7.

τ	0.0	0.5	1.0	2.0	5.0
$K_R(\tau)$	1.753	1.964	2.102	2.321	2.655
$\Lambda(\tau)$	—	0.218	0.174	0.140	0.139
carbon	4.023	4.348	4.700	5.389	6.193

Three corrections follow. (i) Condition M is *false* here—the fossil budget *raises* renewable capacity, which is precisely the channel the condition assumes away. (ii) Λ is *positive*, not negative: the induced build-out really does offset part of the load's footprint, roughly 14–22% of it. The earlier claim that fossil-running "does not even raise K_R" was the barrier speaking. (iii) The *conclusion* is unharmed: $\Lambda < 1$ at every τ, so surplus-only remains the carbon-optimal member of the family, and carbon rises monotonically in τ.

Theorem 4 is therefore sound but its hypothesis is not the operative one: Condition M is sufficient for $\Lambda \le 0$, and what actually holds at interior optima is the weaker and more interesting statement that the offset is partial. The honest economic content is not "the build-out finances nothing" but "the build-out finances about a fifth, and a fifth is not enough."

Remark 8 (What is now proved, and what remains). The leverage question is closed *conditionally*: $\Lambda \le 1$ is a *theorem* given the capacity-monotonicity Condition M, and M is a transparent, checkable statement about the two capacity responses that holds throughout the scanned region. What is not proved is Condition M itself in closed form—signing $K_R'(\tau), K_F'(\tau)$ in the coupled, submodular three-capacity program requires the full inverse Hessian (Week-2's cofactor identity), which we do not carry out. So the residual open step is a clean, isolated comparative-statics sign, not the global optimality claim.

7. Numerical validation

`verify_week4.py` checks, against adaptive quadrature on Beta scale mixtures (symmetric, left- and right-skewed, mildly skewed): the classification identity $\Phi = \Delta$ to $\sim 3\times10^{-8}$ and $\operatorname{sign}\Delta = \operatorname{sign}\Phi$ in every cell (Theorem 1); the partial signs $\partial\Phi/\partial\mu^*_{\text{low}} < 0$, $\partial\Phi/\partial\mu^\circ_{\text{mid}} > 0$ (Proposition 1); the per-mean margins $\mathrm{D}_\ell, \mathrm{D}_m$ ($\Phi_{\text{hi}} = +0.40, +0.49 > 0$) and that the exact *iff* $\mathrm{D}_\ell \Leftrightarrow$ (sweep flips) holds (Theorem 2); and Condition M (K_F non-decreasing, K_R non-increasing along τ) with the resulting $\Lambda(\tau) < 1$ (Theorem 4), with the interior-optimum correction of Remark 7. The suite reports a single pass count.